Phase 3 (part 4): Communities in Graphs

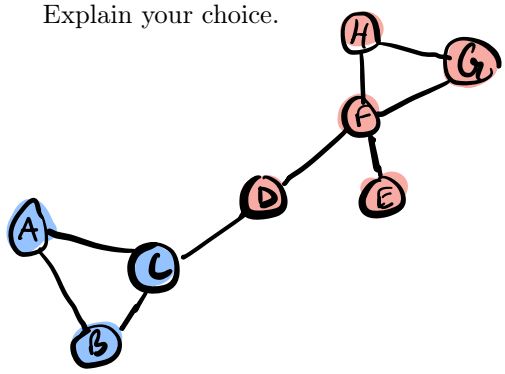
1.] Consider the unweighted, undirected graph on nodes

$$V = \{A, B, C, D, E, F, G, H\}$$

with edges

$$\{AB, AC, BC, CD, DE, DF, EF, FG, GH, FH\}.$$

a.) Draw the graph in Python. Based on the picture alone, propose a community partition of the nodes. Explain your choice.



$C_1 = \{A, B, C\}$

$C_2 = \{D, E, F, G, H\}$

b.) Identify any *bridge-like* edge(s): edges that appear to connect one dense region of the graph to another. Which edge(s) would you remove first if you were trying to split the graph into communities?

Edge (C,D) is a "cut-edge" that splits the graph into communities C_1 and C_2 upon removal.

c.) Remove one bridge-like edge from part (b). What are the connected components of the remaining graph?

G_1: $\{(A,B), (A,C), (B,C)\}$

G_2: $\{(D,E), (D,F), (E,F), (F,G), (G,H), (F,H)\}$

d.) (Python) Build this graph in `networkx`. Compute the edge betweenness centrality of each edge using `nx.edge_betweenness_centrality(G)`.Which edge has the largest edge betweenness?

↳ benerizk_p3ws4_1.py (C,D) has largest edge centrality

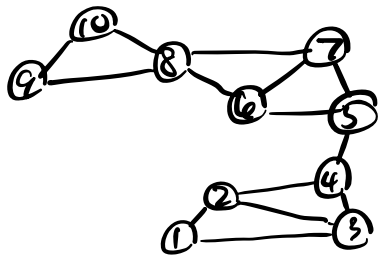
2.] Consider the unweighted, undirected graph on nodes

$$V = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

with edge set

$$E = \{(1,2), (1,3), (2,3), (2,4), (3,4), (4,5), (5,6), (5,7), (6,7), (6,8), (7,8), (8,9), (8,10), (9,10)\}.$$

a.) Draw the graph in Python and propose a partition into communities. (You may propose 2 or 3 communities.) Justify briefly.



$C_1 = \{1,2,3,4\}$

$C_2 = \{5,6,7,8,9,10\}$

b.) Let m be the number of edges, let k_i be the degree of node i, and let A be the adjacency matrix. The *modularity* of a partition is

$m = |E| = 14$

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

$Q = Q_1 + Q_2$

where $\delta(c_i, c_j) = 1$ if nodes i and j are in the same community and 0 otherwise. Compute m and list the degrees k_i for all nodes. Using your proposed partition from part (a), compute Q *by hand* (you may organize the sum by communities).

$k_1 = 2,\ k_2 = 3,\ k_3 = 3,\ k_4 = 3,\ k_5 = 3,\ k_6 = 3,\ k_7 = 3,\ k_8 = 4,\ k_9 = 2,\ k_{10} = 2$

calculations below

$$\begin{cases} Q_1 = \dfrac{154}{784} \\ Q_2 = \dfrac{151}{784} \end{cases} \rightarrow \boxed{Q = \frac{310}{784} \approx 0.3954}$$

c.) Build the graph in `networkx`. Use greedy modularity community detection:

```
from networkx.algorithms.community import greedy_modularity_communities
```

List the communities returned. Compute the modularity of that partition using

```
from networkx.algorithms.community.quality import modularity.
```

Compare your hand partition's modularity to Python's partition modularity.

↳ bemerrick_p3ws4 Python's Communities: $C_1 = \{1,2,3,4\}$, $C_2 = \{5,6,7\}$, $C_3 = \{8,9,10\}$

$Q = 0.446429$

Modularity for $C_1 = \{1,2,3,4\}$

$$Q_1 = \frac{1}{2m}\Big[\left(A_{11} - \frac{k_1k_1}{2m}\right) + \left(A_{12} - \frac{k_1k_2}{2m}\right) + \left(A_{13} - \frac{k_1k_3}{2m}\right) + \left(A_{14} - \frac{k_1k_4}{2m}\right) + \cdots$$

$$\left(A_{21} - \frac{k_2k_1}{2m}\right) + \left(A_{22} - \frac{k_2k_2}{2m}\right) + \left(A_{23} - \frac{k_2k_3}{2m}\right) + \left(A_{24} - \frac{k_2k_4}{2m}\right) + \cdots$$

$$\left(A_{31} - \frac{k_3k_1}{2m}\right) + \left(A_{32} - \frac{k_3k_2}{2m}\right) + \left(A_{33} - \frac{k_3k_3}{2m}\right) + \left(A_{34} - \frac{k_3k_4}{2m}\right) + \cdots$$

$$\left(A_{41} - \frac{k_4k_1}{2m}\right) + \left(A_{42} - \frac{k_4k_2}{2m}\right) + \left(A_{43} - \frac{k_4k_3}{2m}\right) + \left(A_{44} - \frac{k_4k_4}{2m}\right)\Big]$$

$$= \frac{1}{28}\Big[\left(0 - \frac{4}{28}\right) + 2\left(1 - \frac{6}{28}\right) + 2\left(1 - \frac{6}{28}\right) + 2\left(0 - \frac{6}{28}\right) + \left(0 - \frac{9}{28}\right) + \cdots$$

$$2\left(1 - \frac{9}{28}\right) + 2\left(1 - \frac{9}{28}\right) + \left(0 - \frac{9}{28}\right) + 2\left(1 - \frac{9}{28}\right) + \left(0 - \frac{9}{28}\right)\Big]$$

$$= \frac{1}{28^2}\Big[-4 + 44 + 44 - 12 - 9 + 38 + 38 - 9 + 38 - 9\Big]$$

$$\boxed{Q_1 = \frac{159}{784}}$$

Modularity for $C_2 = \{5,6,7,8,9,10\}$

$$
\begin{aligned}
Q_2 = \frac{1}{2m}\Big[& \left(A_{55}-\frac{k_5k_5}{2m}\right)+\left(A_{56}-\frac{k_5k_6}{2m}\right)+\left(A_{57}-\frac{k_5k_7}{2m}\right)+\left(A_{58}-\frac{k_5k_8}{2m}\right)+\left(A_{59}-\frac{k_5k_9}{2m}\right)+\left(A_{5,10}-\frac{k_5k_{10}}{2m}\right)+\cdots \\
& \left(A_{65}-\frac{k_6k_5}{2m}\right)+\left(A_{66}-\frac{k_6k_6}{2m}\right)+\left(A_{67}-\frac{k_6k_7}{2m}\right)+\left(A_{68}-\frac{k_6k_8}{2m}\right)+\left(A_{69}-\frac{k_6k_9}{2m}\right)+\left(A_{6,10}-\frac{k_6k_{10}}{2m}\right)+\cdots \\
& \left(A_{75}-\frac{k_7k_5}{2m}\right)+\left(A_{76}-\frac{k_7k_6}{2m}\right)+\left(A_{77}-\frac{k_7k_7}{2m}\right)+\left(A_{78}-\frac{k_7k_8}{2m}\right)+\left(A_{79}-\frac{k_7k_9}{2m}\right)+\left(A_{7,10}-\frac{k_7k_{10}}{2m}\right)+\cdots \\
& \left(A_{85}-\frac{k_8k_5}{2m}\right)+\left(A_{86}-\frac{k_8k_6}{2m}\right)+\left(A_{87}-\frac{k_8k_7}{2m}\right)+\left(A_{88}-\frac{k_8k_8}{2m}\right)+\left(A_{89}-\frac{k_8k_9}{2m}\right)+\left(A_{8,10}-\frac{k_8k_{10}}{2m}\right)+\cdots \\
& \left(A_{95}-\frac{k_9k_5}{2m}\right)+\left(A_{96}-\frac{k_9k_6}{2m}\right)+\left(A_{97}-\frac{k_9k_7}{2m}\right)+\left(A_{98}-\frac{k_9k_8}{2m}\right)+\left(A_{99}-\frac{k_9k_9}{2m}\right)+\left(A_{9,10}-\frac{k_9k_{10}}{2m}\right)+\cdots \\
& \left(A_{10,5}-\frac{k_{10}k_5}{2m}\right)+\left(A_{10,6}-\frac{k_{10}k_6}{2m}\right)+\left(A_{10,7}-\frac{k_{10}k_7}{2m}\right)+\left(A_{10,8}-\frac{k_{10}k_8}{2m}\right)+\left(A_{10,9}-\frac{k_{10}k_9}{2m}\right)+\left(A_{10,10}-\frac{k\,k}{2m}\right)\Big]
\end{aligned}
$$

$$
\begin{aligned}
= \frac{1}{28}\Big[& \left(0-\frac{9}{28}\right)+2\left(1-\frac{9}{28}\right)+2\left(1-\frac{9}{28}\right)+2\left(0-\frac{12}{28}\right)+2\left(0-\frac{6}{28}\right)+2\left(0-\frac{6}{28}\right)+\cdots \\
& \left(0-\frac{9}{28}\right)+2\left(1-\frac{9}{28}\right)+2\left(1-\frac{12}{28}\right)+2\left(0-\frac{6}{28}\right)+2\left(0-\frac{6}{28}\right)+\cdots \\
& +\left(0-\frac{9}{28}\right)+2\left(1-\frac{12}{28}\right)+2\left(0-\frac{6}{28}\right)+2\left(0-\frac{6}{28}\right)+\cdots \\
& +\left(0-\frac{16}{28}\right)+2\left(1-\frac{8}{28}\right)+2\left(1-\frac{8}{28}\right)+\cdots \\
& +\left(0-\frac{4}{28}\right)+2\left(1-\frac{4}{28}\right)+\cdots \\
& +\left(0-\frac{4}{28}\right)\Big]
\end{aligned}
$$

$$
\begin{aligned}
= \frac{1}{28^2}\Big[& -9+38+38-24-12-12+\cdots \\
& -9+38+28-12-12+\cdots \\
& -9+28-12-12+\cdots \\
& -16+40+40+\cdots \\
& -4+48+\cdots \\
& -4\Big]
\end{aligned}
$$

$$\boxed{Q_2 = \frac{151}{784}}$$